

MRSIPrep report: SynthMRSI-Project · sub-01 ses-01

Inputs

BIDS directory: /data

Output directory: /out/mrsiprep

Parcellation mode: chimera

Tissue backend: synthseg-fast

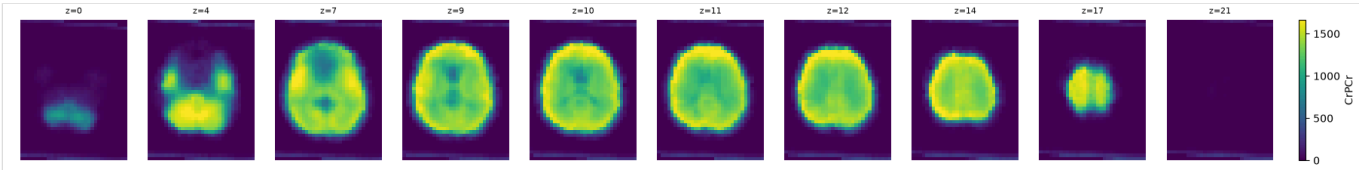
MRSI Raw QC

MRSI Raw QC

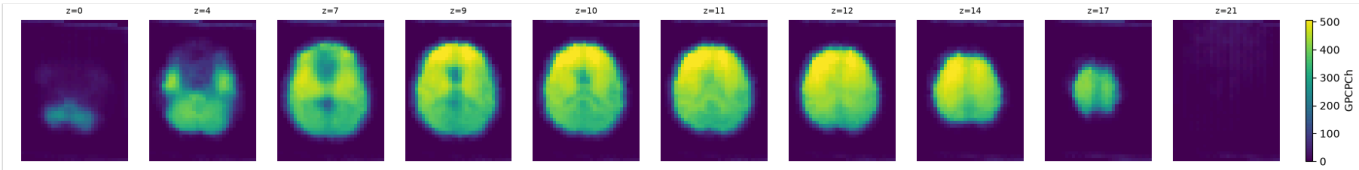
metabolite	n_total_voxels	mean_snr	median_snr	mean_linewidth	median_linewidth	mean_crlb	median_crlb
NAANAAG	20738	14.135942	15.0	0.038052	0.036	4.933007	4.0
GPCPCh	20738	14.228251	15.0	0.037878	0.036	6.853189	6.0
CrPCr	20738	14.257129	15.0	0.037799	0.036	6.431937	5.0
GluGln	20738	14.345459	15.0	0.037970	0.036	11.069957	11.0
Ins	20738	14.465190	15.0	0.037318	0.033	8.926435	8.0

Raw metabolite maps (pre-pipeline)

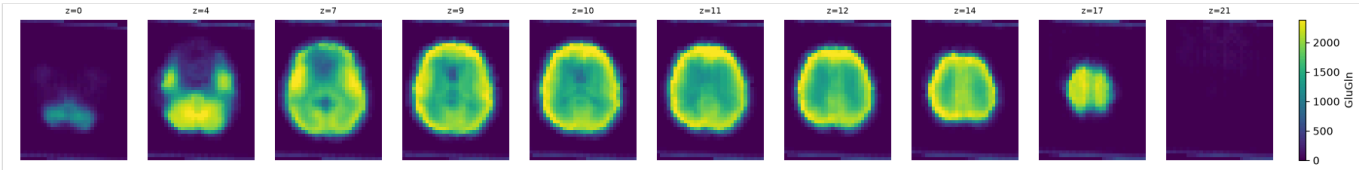
Metabolite: CrPCr



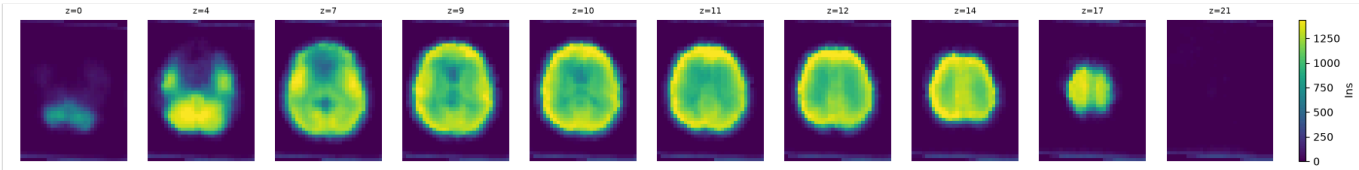
Metabolite: GPCPCh



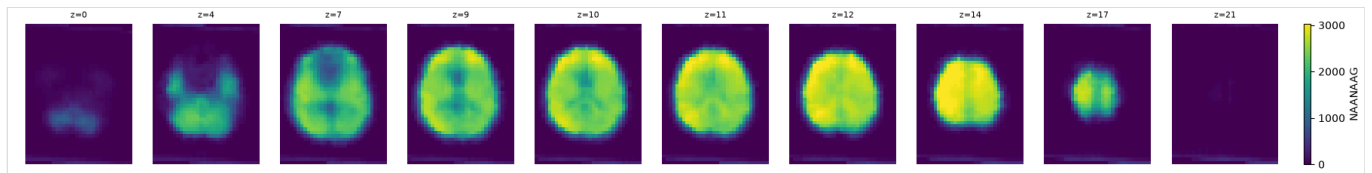
Metabolite: GluGln



Metabolite: Ins

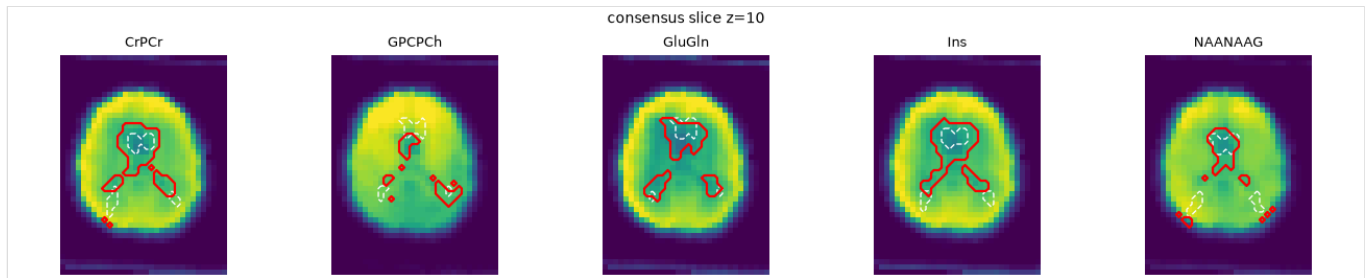


Metabolite: NAANAAG



Ventricle visibility (pre-coregistration)

— detected ventricle --- MNI prior native MRSI space, pre-coregistration · all metabolites at z=10



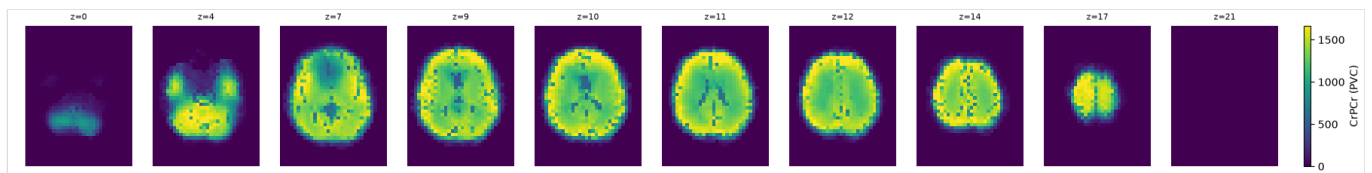
MRSI PVC

MRSI PVC

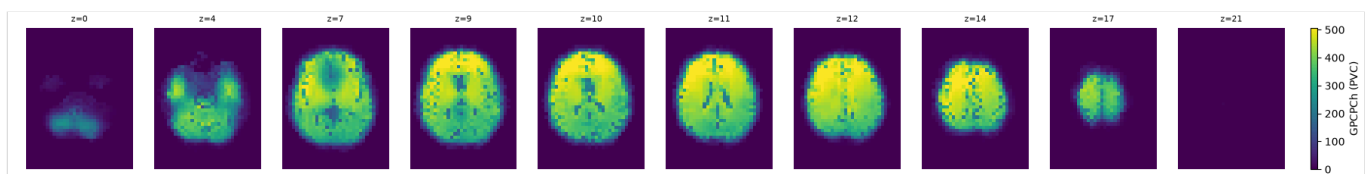
Partial-volume correction

Partial-volume-corrected metabolite maps, on the native MRSI grid, at the same slices as the MRSI Raw QC tab so the two can be compared directly.

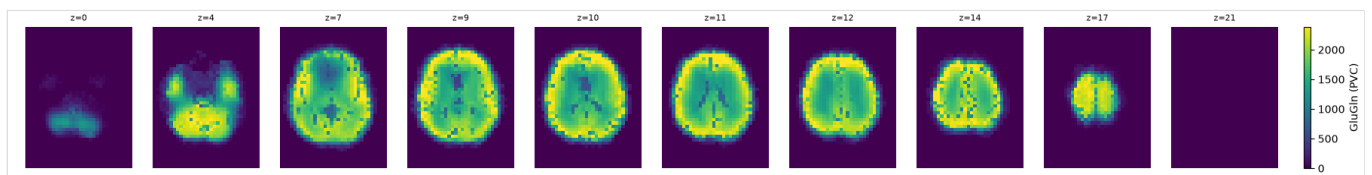
Metabolite: CrPCr



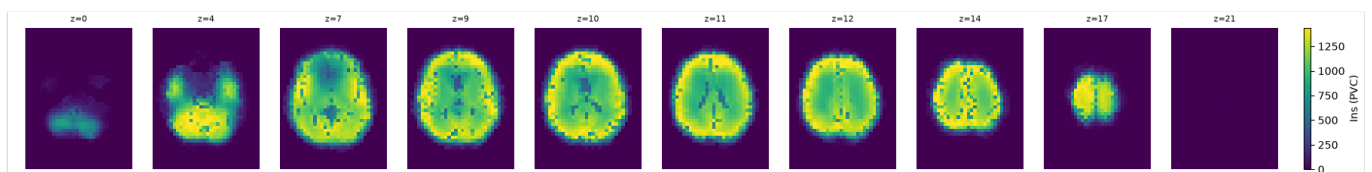
Metabolite: GPCPCh



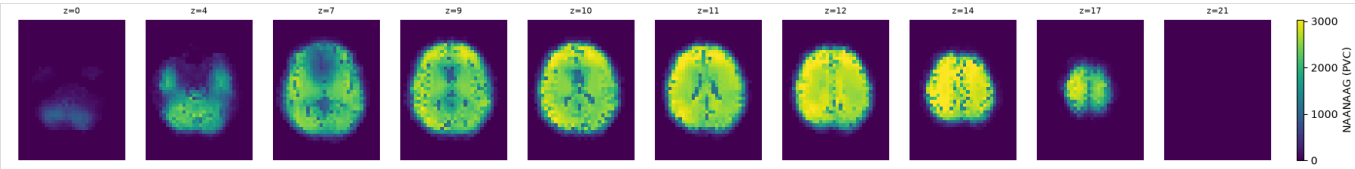
Metabolite: GluGln



Metabolite: Ins



Metabolite: NAANAAG



Spike Filter

Spike Filter

Filtering summary

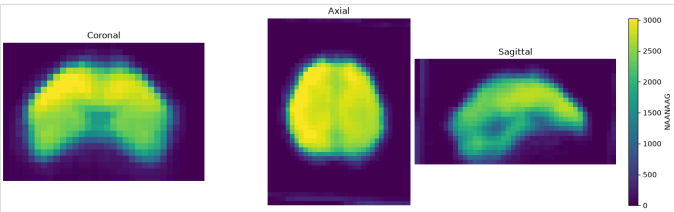
Slices centered on the centroid of all voxels repaired by spike/missing-voxel filtering, across all metabolites.

Metabolite	Spike voxels detected
NAANAAG	0
GPCPCh	0
CrPCr	7
GluGln	25
Ins	20

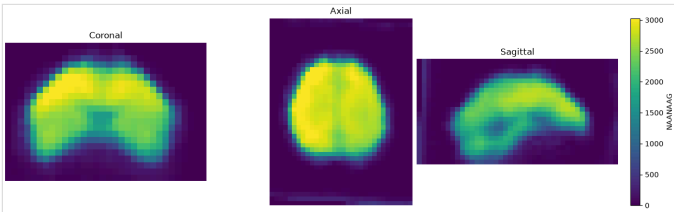
Metabolite: NAANAAG

Spike voxels detected: 0

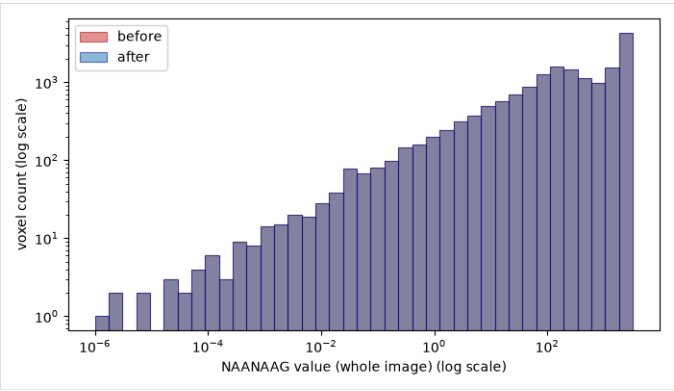
Before



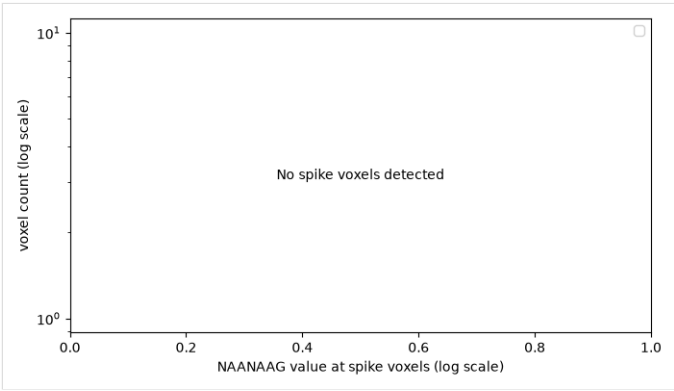
After



Whole image



Spike voxels only

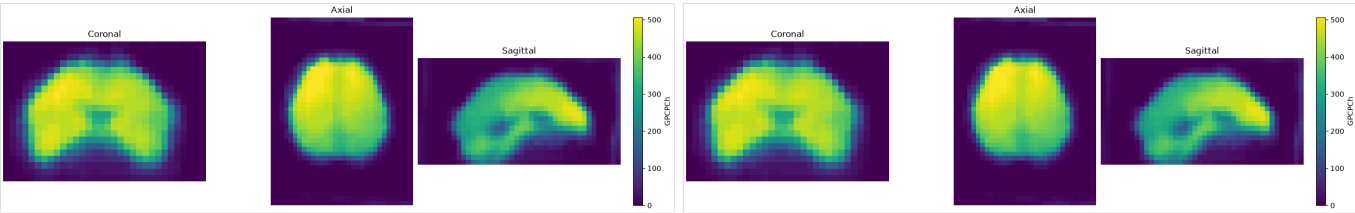


Metabolite: GPCPCh

Spike voxels detected: 0

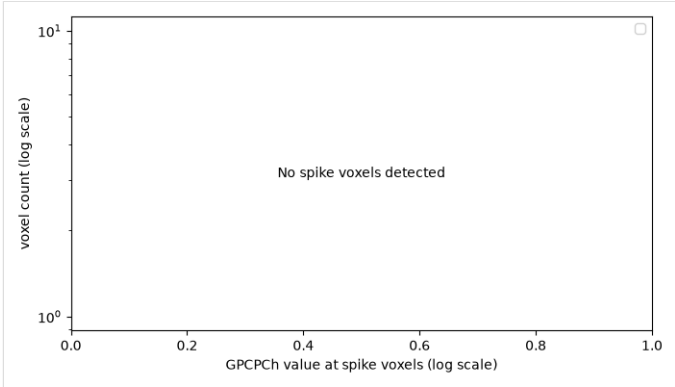
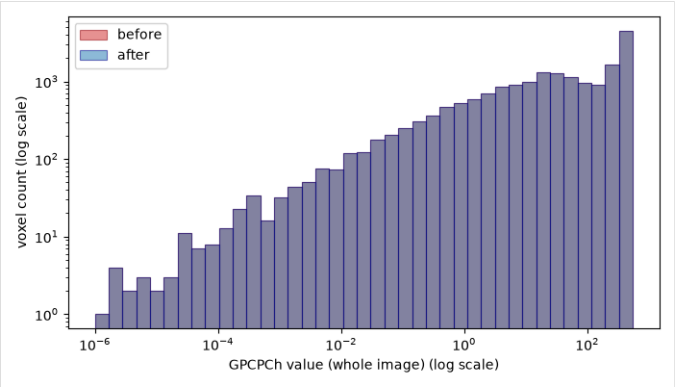
Before

After



Whole image

Spike voxels only

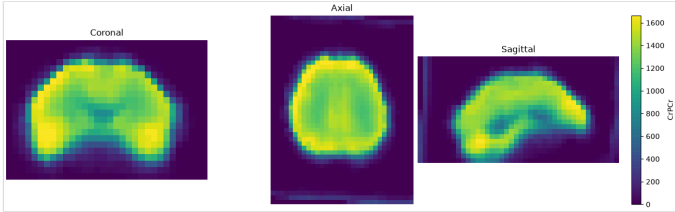
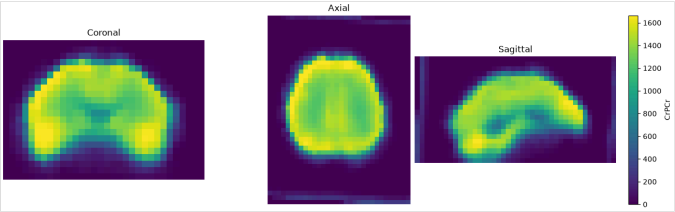


Metabolite: CrPCr

Spike voxels detected: 7

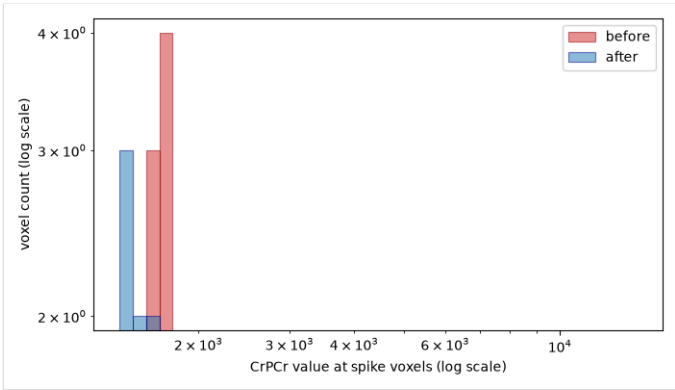
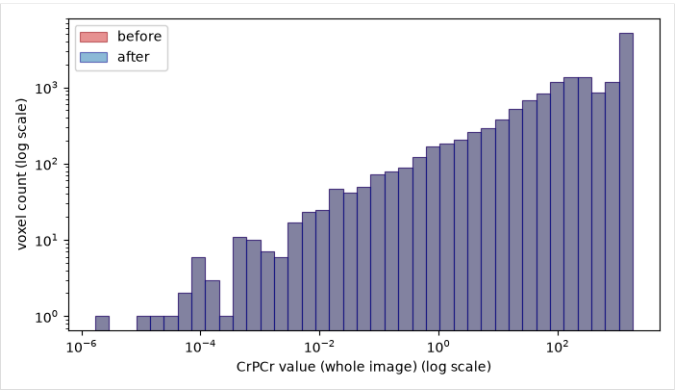
Before

After



Whole image

Spike voxels only

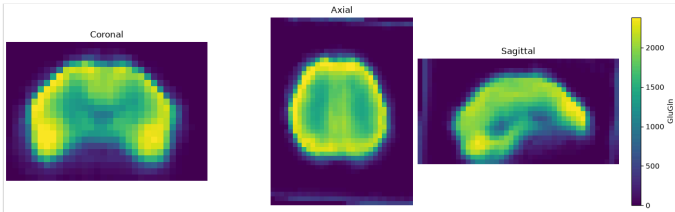
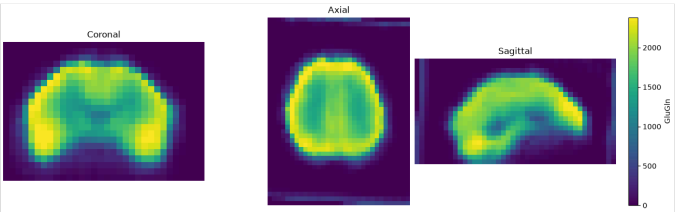


Metabolite: GluGln

Spike voxels detected: 25

Before

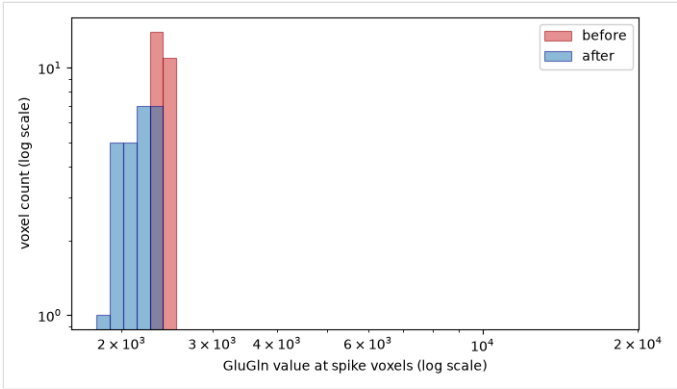
After



Whole image



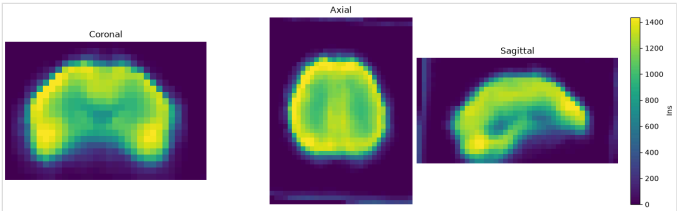
Spike voxels only



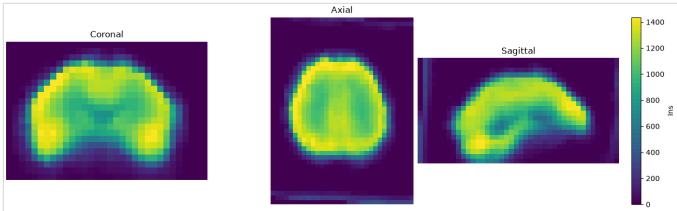
Metabolite: Ins

Spike voxels detected: 20

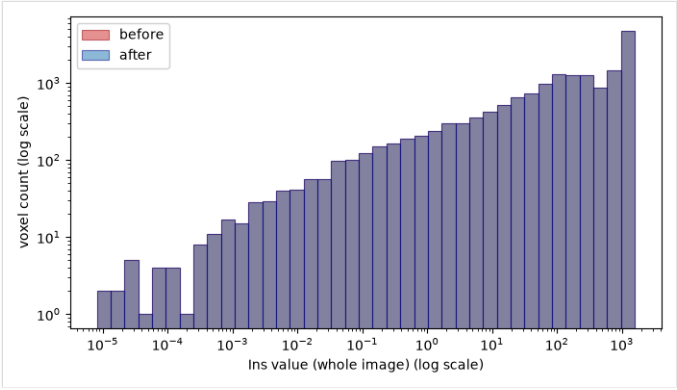
Before



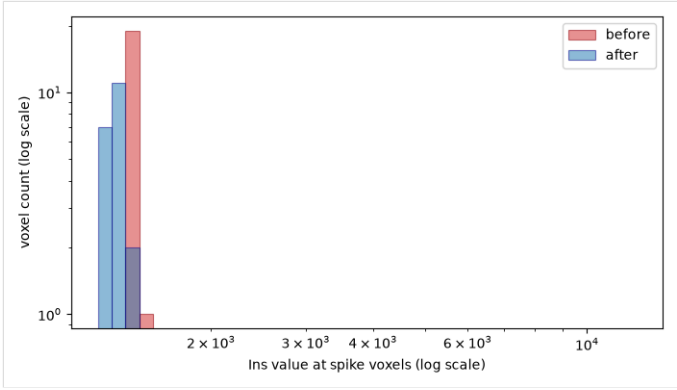
After



Whole image



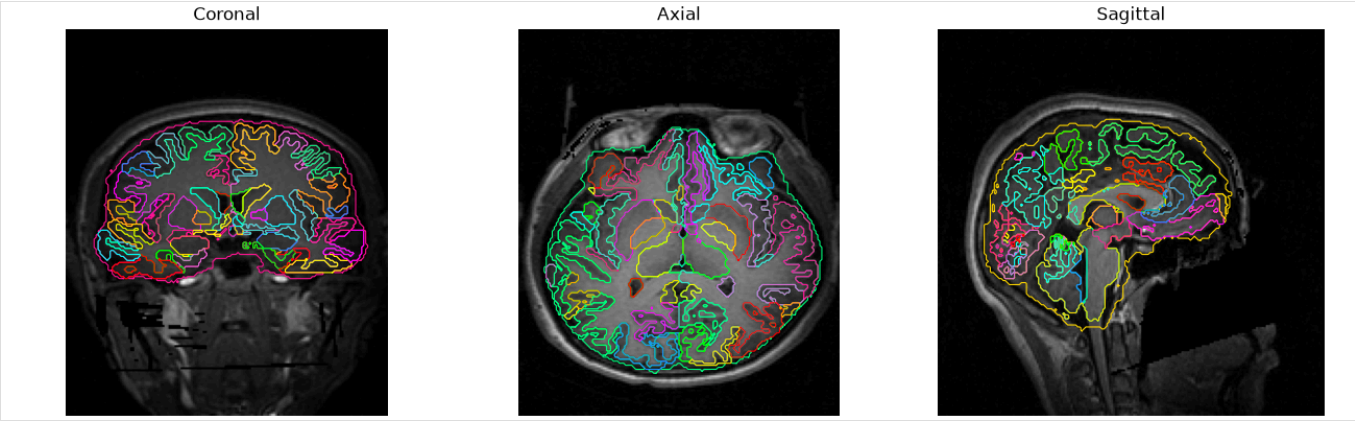
Spike voxels only



Anatomical

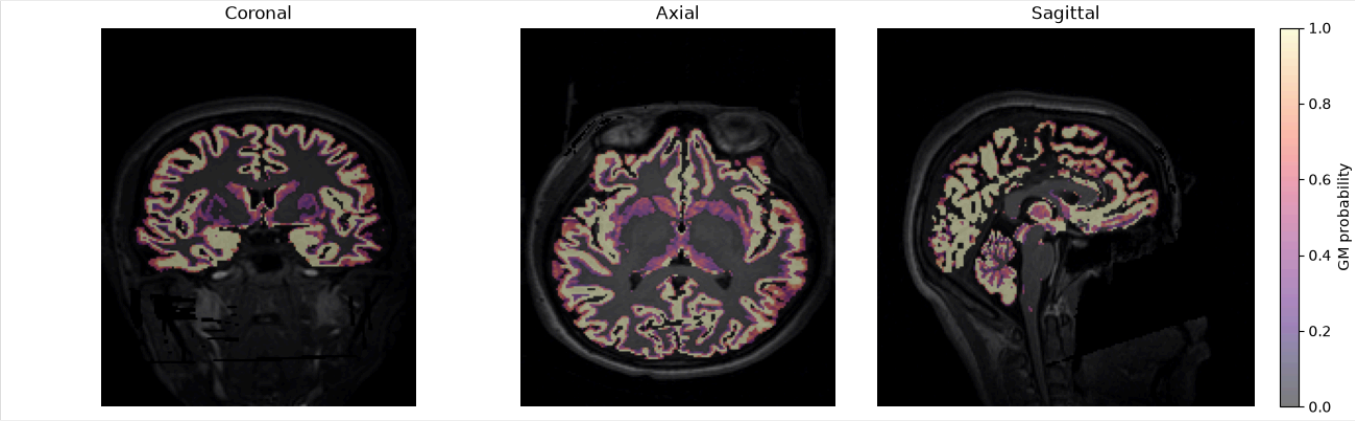
Anatomical

Tissue label outlines

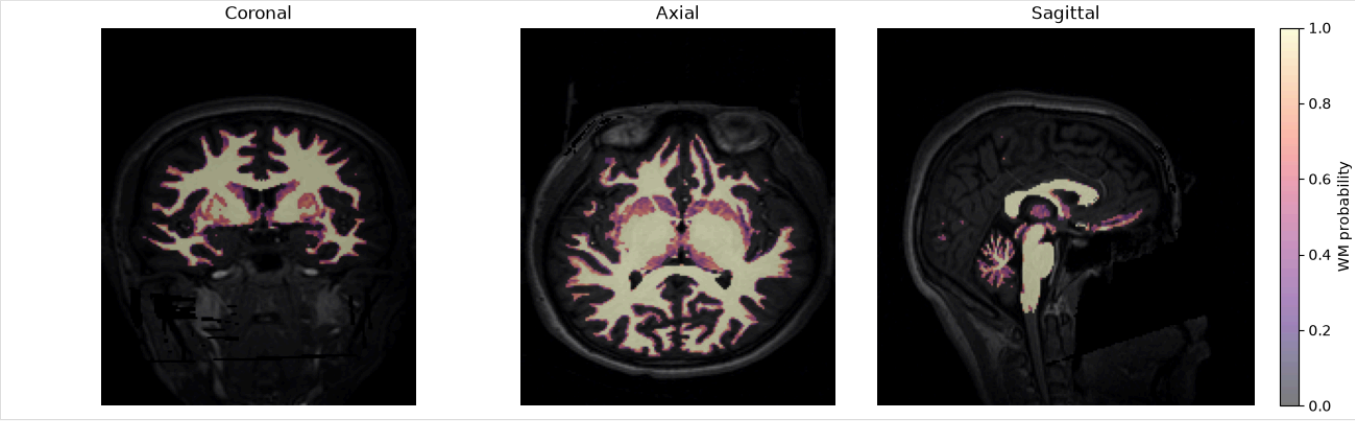


Tissue probability maps

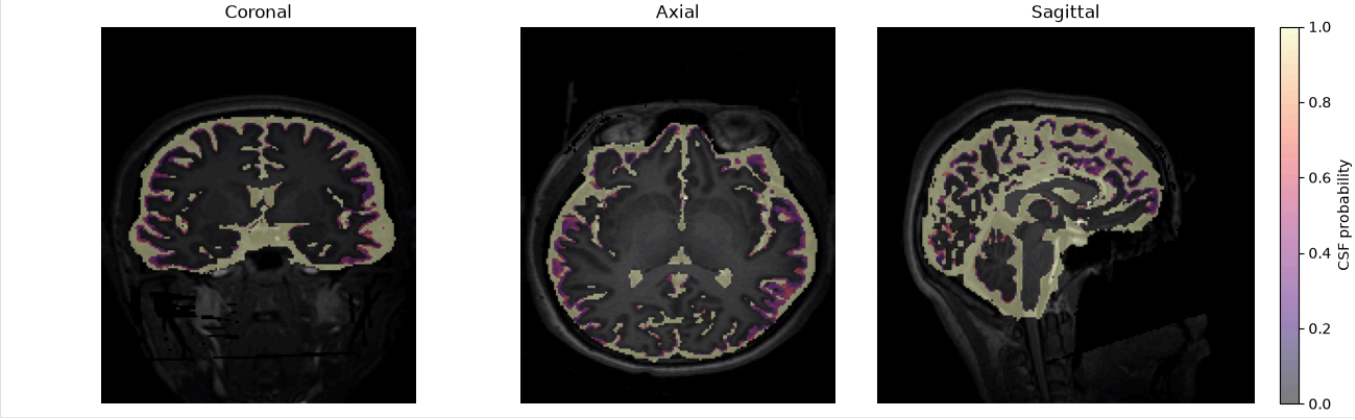
GM



WM



CSF

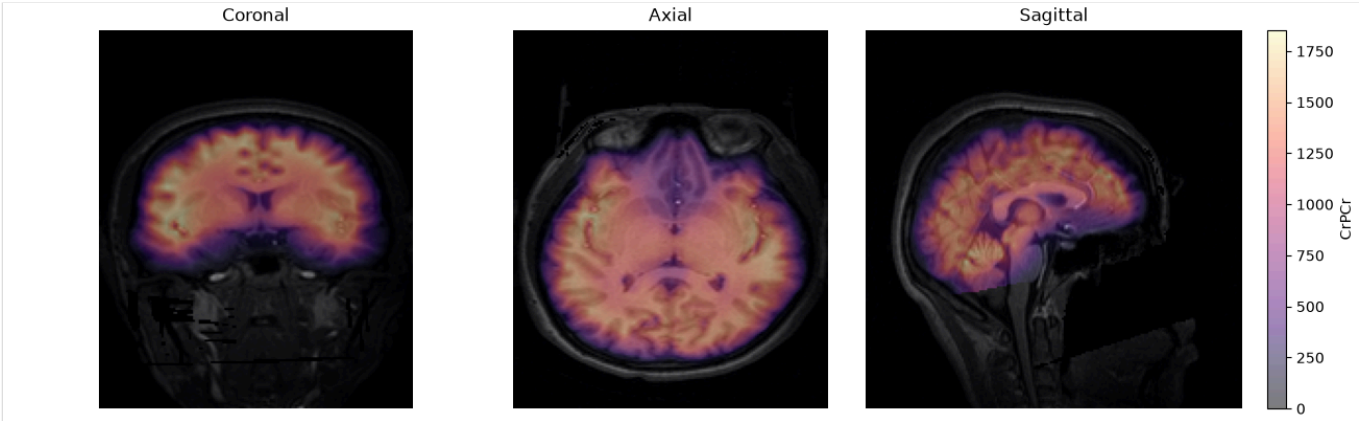


T1-space alignment

T1-space alignment

T1w-space alignment

Reference metabolite map (CrPCr) overlaid on raw T1w.



Signal-weighted leakage

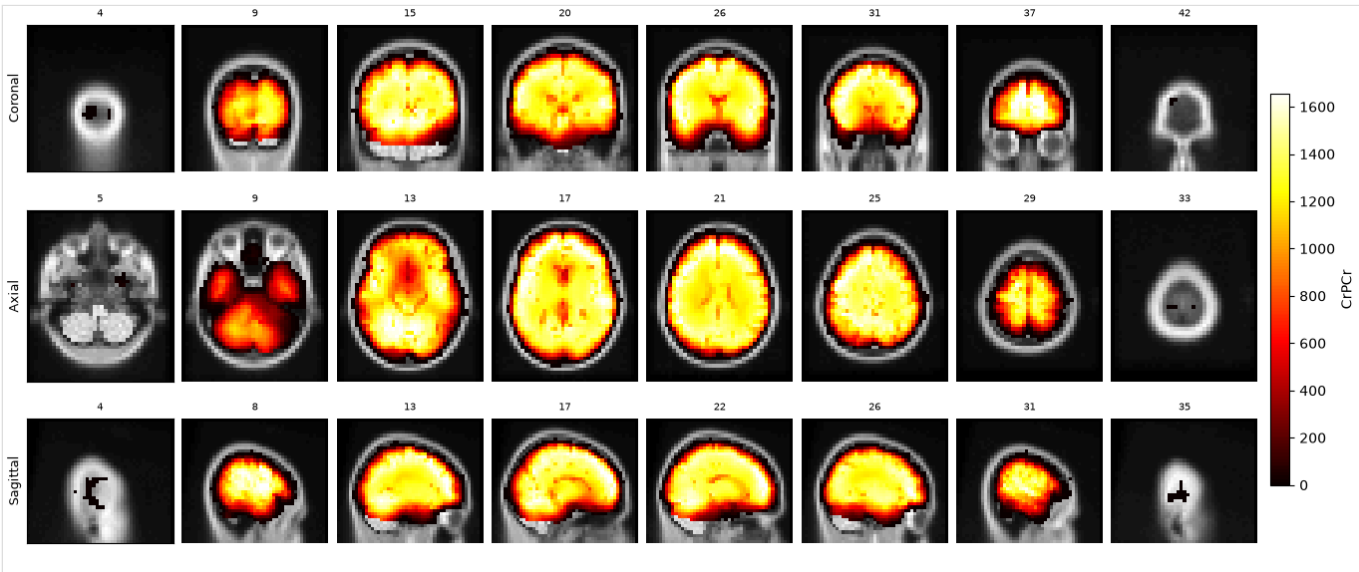
Not computed: T1w-space leakage requires the full T1w-space resampled metabolite maps (`--output-mrsi-t1w`), which weren't requested for this run. The alignment image above uses a separate, lightweight report-only resampling and doesn't imply this data is available.

Template-space alignment

Template-space alignment

Template-space alignment

Reference metabolite map (CrPCr) overlaid on the full-head MNI152NLin2009cAsym template at 5 mm.



Signal-weighted leakage

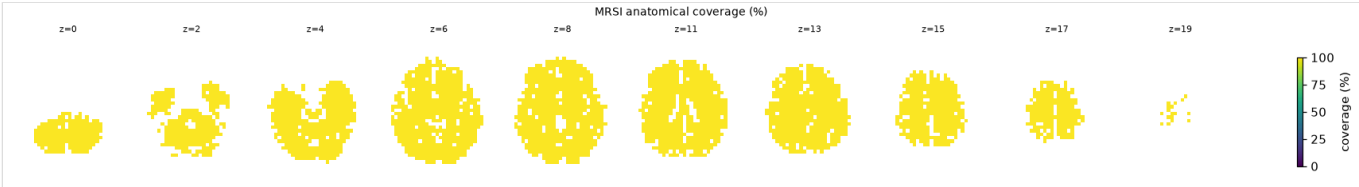
metabolite	n_quality_voxels	total_signal_mass	leakage_signal_mass	leakage_fraction	leakage_percent
CrPCr	73752	15010842.000	89485.297	0.006	0.596
GPCPCh	73875	4183460.750	29170.521	0.007	0.697
GluGln	73625	20428784.000	147303.656	0.007	0.721

metabolite	n_quality_voxels	total_signal_mass	leakage_signal_mass	leakage_fraction	leakage_percent
Ins	73403	12725528.000	87774.406	0.007	0.690
NAANAAG	74278	23589286.000	149212.031	0.006	0.633

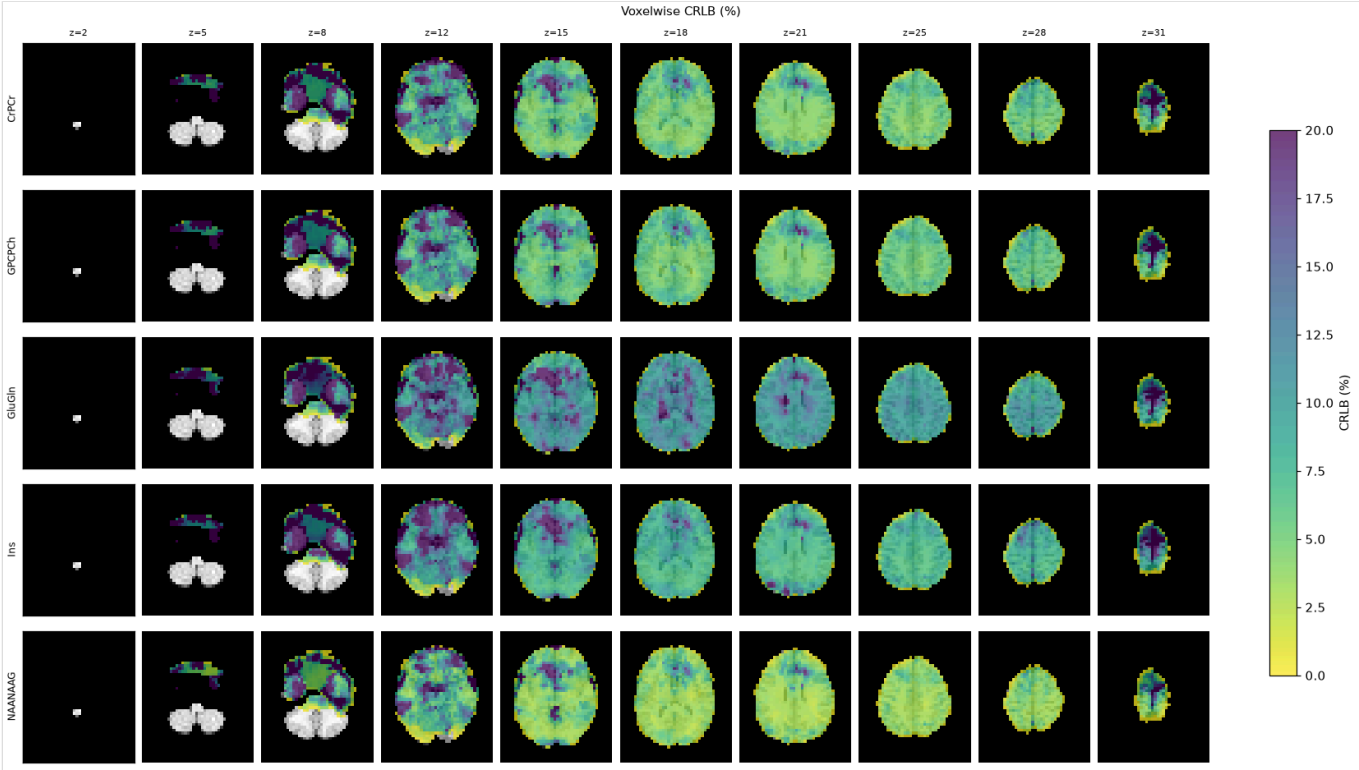
Coverage

Coverage

MRSI anatomical coverage



Parcelwise CRLB quality (green reliable / red unreliable)



qc_valid_fraction is the fraction of a parcel's MRSI voxels that passed every per-voxel quality threshold in force for this run (CRLB $\leq$ 20.0%, SNR $\geq$ 4.0, linewidth $\leq$ 0.1), averaged over metabolites. Rows are ordered worst first.

parcel_id †	parcel_name †	hemisphere †	mean_crlb †	qc_valid_fraction †
2006	ctx-rh-entorhinal	R	26.900	0.000
2032	ctx-rh-frontalpole	R	417.267	0.143
1006	ctx-lh-entorhinal	L	163.415	0.169
46	Right-Cerebellum-White-Matter	R	16.588	0.274
2033	ctx-rh-temporalpole	R	155.200	0.329
1033	ctx-lh-temporalpole	L	119.692	0.338
18	Left-Amygdala	L	22.000	0.345
47	Right-Cerebellum-Cortex	R	31.257	0.367
7	Left-Cerebellum-White-Matter	L	13.315	0.390
8	Left-Cerebellum-Cortex	L	14.707	0.422
2019	ctx-rh-parsorbitalis	R	28.176	0.459

parcel_id :	parcel_name :	hemisphere :	mean_crlb :	qc_valid_fraction :
1009	ctx-lh-inferiortemporal	L	35.404	0.466
1032	ctx-lh-frontalpole	L	115.267	0.467
16	Brain-Stem	NaN	17.059	0.512
2011	ctx-rh-lateraloccipital	R	7.023	0.564
1027	ctx-lh-rostralmiddlefrontal	L	8.116	0.570
26	Left-Accumbens-area	L	89.433	0.633
1011	ctx-lh-lateraloccipital	L	7.805	0.655
2027	ctx-rh-rostralmiddlefrontal	R	15.281	0.686
2009	ctx-rh-inferiortemporal	R	28.685	0.737
1003	ctx-lh-caudalmiddlefrontal	L	6.733	0.750
1015	ctx-lh-middletemporal	L	24.200	0.759
1007	ctx-lh-fusiform	L	24.515	0.775
1012	ctx-lh-lateralorbitofrontal	L	15.264	0.783
2012	ctx-rh-lateralorbitofrontal	R	22.735	0.796
17	Left-Hippocampus	L	27.143	0.814
2026	ctx-rh-rostralanteriorcingulate	R	13.333	0.833
1028	ctx-lh-superiorfrontal	L	7.959	0.842
1016	ctx-lh-parahippocampal	L	12.114	0.843
60	Right-VentralDC	R	11.329	0.858
1022	ctx-lh-postcentral	L	6.559	0.863
53	Right-Hippocampus	R	27.496	0.864
1026	ctx-lh-rostralanteriorcingulate	L	29.400	0.873
54	Right-Amygdala	R	11.675	0.875
2007	ctx-rh-fusiform	R	17.431	0.897
1014	ctx-lh-medialorbitofrontal	L	24.898	0.902
1031	ctx-lh-supramarginal	L	6.697	0.904
50	Right-Caudate	R	10.030	0.910
11	Left-Caudate	L	10.021	0.916
2003	ctx-rh-caudalmiddlefrontal	R	7.645	0.917
2002	ctx-rh-caudalanteriorcingulate	R	11.283	0.917
2014	ctx-rh-medialorbitofrontal	R	15.082	0.918
2020	ctx-rh-parstriangularis	R	9.707	0.919
13	Left-Pallidum	L	11.500	0.920
1019	ctx-lh-parsorbitalis	L	7.117	0.923
1013	ctx-lh-lingual	L	7.418	0.925
2028	ctx-rh-superiorfrontal	R	7.425	0.926
2022	ctx-rh-postcentral	R	6.719	0.929
2024	ctx-rh-precentral	R	6.782	0.931
2031	ctx-rh-supramarginal	R	6.269	0.931
12	Left-Putamen	L	8.051	0.937
28	Left-VentralDC	L	13.208	0.938
2029	ctx-rh-superiorparietal	R	6.944	0.939
2015	ctx-rh-middletemporal	R	8.852	0.950
1024	ctx-lh-precentral	L	7.028	0.950
58	Right-Accumbens-area	R	14.400	0.950
1029	ctx-lh-superiorparietal	L	6.952	0.951
1008	ctx-lh-inferiorparietal	L	7.000	0.952
2	Left-Cerebral-White-Matter	L	9.650	0.953
1002	ctx-lh-caudalanteriorcingulate	L	9.525	0.958

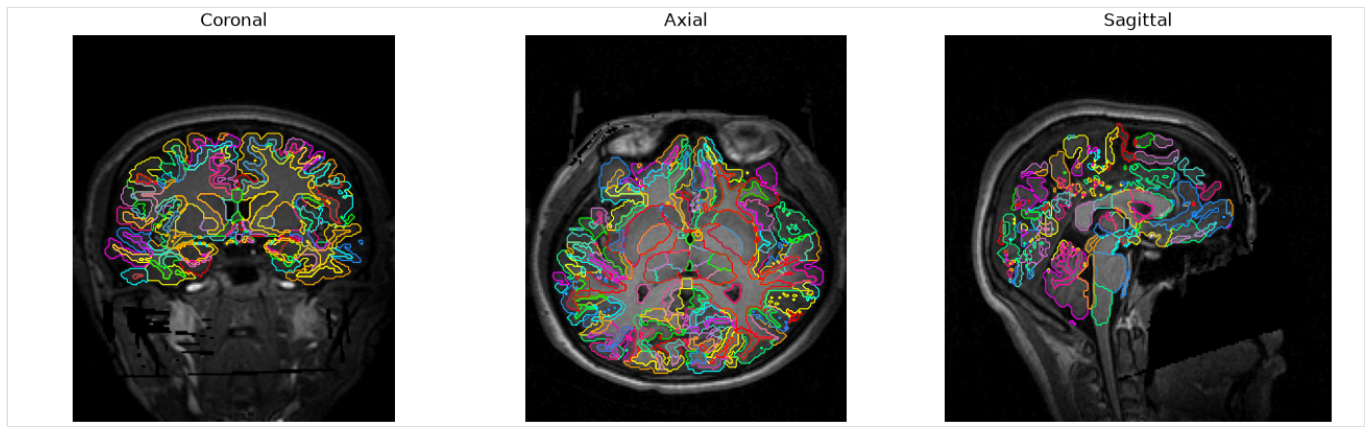
parcel_id :	parcel_name :	hemisphere :	mean_crlb :	qc_valid_fraction :
1030	ctx-lh-superiortemporal	L	7.745	0.960
2013	ctx-rh-lingual	R	7.119	0.964
1017	ctx-lh-paracentral	L	7.353	0.967
1035	ctx-lh-insula	L	7.156	0.968
2017	ctx-rh-paracentral	R	7.119	0.969
2016	ctx-rh-parahippocampal	R	8.886	0.971
41	Right-Cerebral-White-Matter	R	8.019	0.972
2035	ctx-rh-insula	R	7.508	0.973
51	Right-Putamen	R	7.677	0.974
1018	ctx-lh-parsopercularis	L	7.141	0.976
52	Right-Pallidum	R	9.667	0.978
1025	ctx-lh-precuneus	L	6.205	0.984
1020	ctx-lh-parstriangularis	L	7.825	0.992
10	Left-Thalamus	L	7.504	0.992
49	Right-Thalamus	R	6.950	0.996
2030	ctx-rh-superiortemporal	R	6.786	0.997
2008	ctx-rh-inferiorparietal	R	6.339	0.998
1001	ctx-lh-bankssts	L	5.788	1.000
1005	ctx-lh-cuneus	L	6.938	1.000
1010	ctx-lh-isthmuscingulate	L	5.848	1.000
2005	ctx-rh-cuneus	R	6.965	1.000
2001	ctx-rh-bankssts	R	6.306	1.000
1034	ctx-lh-transversetemporal	L	6.000	1.000
1023	ctx-lh-posteriorcingulate	L	5.933	1.000
1021	ctx-lh-pericalcarine	L	7.600	1.000
2021	ctx-rh-pericalcarine	R	6.073	1.000
2018	ctx-rh-parsopercularis	R	7.993	1.000
2010	ctx-rh-isthmuscingulate	R	5.985	1.000
2023	ctx-rh-posteriorcingulate	R	5.936	1.000
2025	ctx-rh-precuneus	R	5.789	1.000
2034	ctx-rh-transversetemporal	R	5.691	1.000

Parcellation

Parcellation

Parcellation outlines (T1w space)

506 regions.



Regional metabolites: chimeraLFMIHIFIFF-scale3

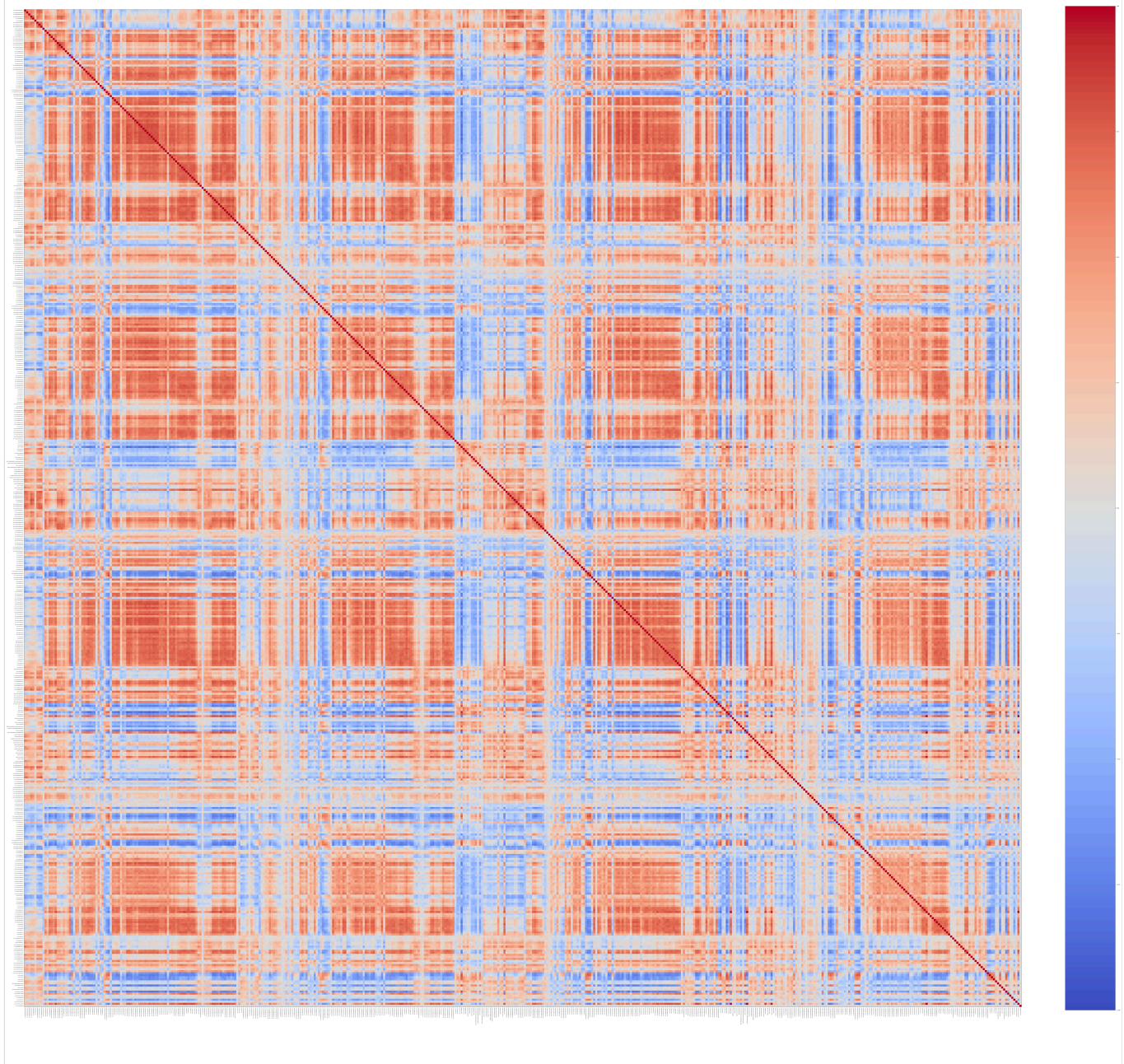
subject	session	atlas	scale	parcel_id	parcel_name	hemisphere	metabolite	mean	median
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	1	ctx-rh-lateralorbitofrontal_1	R	NAANAAG	630.298828	456.964752
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	1	ctx-rh-lateralorbitofrontal_1	R	GPCPCh	195.250153	170.521088
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	1	ctx-rh-lateralorbitofrontal_1	R	CrPCr	518.341553	445.490265
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	1	ctx-rh-lateralorbitofrontal_1	R	GluGln	699.686279	621.933167
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	1	ctx-rh-lateralorbitofrontal_1	R	Ins	473.752930	422.292358
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	2	ctx-rh-lateralorbitofrontal_2	R	NAANAAG	1244.026001	1067.720703
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	2	ctx-rh-lateralorbitofrontal_2	R	GPCPCh	300.121735	282.140564
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	2	ctx-rh-lateralorbitofrontal_2	R	CrPCr	967.416504	927.080566
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	2	ctx-rh-lateralorbitofrontal_2	R	GluGln	1381.926025	1334.387939
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	2	ctx-rh-lateralorbitofrontal_2	R	Ins	815.326965	792.523193
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	3	ctx-rh-lateralorbitofrontal_3	R	NAANAAG	621.122253	257.051270
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	3	ctx-rh-lateralorbitofrontal_3	R	GPCPCh	192.981369	102.995895
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	3	ctx-rh-lateralorbitofrontal_3	R	CrPCr	697.192932	697.192932
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	3	ctx-rh-lateralorbitofrontal_3	R	GluGln	806.283264	486.878296
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	3	ctx-rh-lateralorbitofrontal_3	R	Ins	542.843933	542.843933
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	4	ctx-rh-lateralorbitofrontal_4	R	NAANAAG	734.847595	637.379211
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	4	ctx-rh-lateralorbitofrontal_4	R	GPCPCh	206.235001	199.953369
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	4	ctx-rh-lateralorbitofrontal_4	R	CrPCr	636.945801	601.967957
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	4	ctx-rh-lateralorbitofrontal_4	R	GluGln	1184.965576	1151.621460
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	4	ctx-rh-lateralorbitofrontal_4	R	Ins	525.538757	498.606323
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	5	ctx-rh-parsorbitalis_1	R	NAANAAG	1144.217407	1240.765381
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	5	ctx-rh-parsorbitalis_1	R	GPCPCh	283.328186	305.798828

subject	session	atlas	scale	parcel_id	parcel_name	hemisphere	metabolite	mean	median
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	5	ctx-rh-parsorbitalis_1	R	CrPCr	935.324890	1065.844482
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	5	ctx-rh-parsorbitalis_1	R	GluGln	1382.558716	1575.562378
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	5	ctx-rh-parsorbitalis_1	R	Ins	893.044312	869.093872
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	6	ctx-rh-frontalpole_1	R	NAANAAG	216.626358	216.626358
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	6	ctx-rh-frontalpole_1	R	GPCPCh	NaN	NaN
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	6	ctx-rh-frontalpole_1	R	CrPCr	NaN	NaN
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	6	ctx-rh-frontalpole_1	R	GluGln	346.515808	346.515808
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	6	ctx-rh-frontalpole_1	R	Ins	NaN	NaN
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	7	ctx-rh-medialorbitofrontal_1	R	NAANAAG	1281.047485	1128.151001
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	7	ctx-rh-medialorbitofrontal_1	R	GPCPCh	324.675171	338.032745
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	7	ctx-rh-medialorbitofrontal_1	R	CrPCr	1019.077332	980.421326
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	7	ctx-rh-medialorbitofrontal_1	R	GluGln	1585.049683	1573.024902
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	7	ctx-rh-medialorbitofrontal_1	R	Ins	865.067139	828.930237
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	8	ctx-rh-medialorbitofrontal_2	R	NAANAAG	388.223907	288.539520
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	8	ctx-rh-medialorbitofrontal_2	R	GPCPCh	155.074188	141.352188
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	8	ctx-rh-medialorbitofrontal_2	R	CrPCr	377.013489	320.113159
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	8	ctx-rh-medialorbitofrontal_2	R	GluGln	517.571716	430.455475
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	8	ctx-rh-medialorbitofrontal_2	R	Ins	333.213348	292.206116
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	9	ctx-rh-medialorbitofrontal_3	R	NAANAAG	450.511658	381.321075
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	9	ctx-rh-medialorbitofrontal_3	R	GPCPCh	159.792419	149.397308
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	9	ctx-rh-medialorbitofrontal_3	R	CrPCr	430.992737	389.962280
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	9	ctx-rh-medialorbitofrontal_3	R	GluGln	534.949768	513.074280
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	9	ctx-rh-medialorbitofrontal_3	R	Ins	362.288910	344.122375
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	10	ctx-rh-parstriangularis_1	R	NAANAAG	2050.917969	2026.731079
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	10	ctx-rh-parstriangularis_1	R	GPCPCh	386.035248	384.524506
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	10	ctx-rh-parstriangularis_1	R	CrPCr	1233.269165	1260.648682
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	10	ctx-rh-parstriangularis_1	R	GluGln	1754.313110	1812.510498
sub-01	ses-01	chimeraLFMIHIFIFF	scale3	10	ctx-rh-parstriangularis_1	R	Ins	1035.241089	1057.267578

Connectivity

Connectivity

Connectivity matrix (spearman)



MRSinMRS

MRSinMRS

Parameter	Value	Unit
AcquisitionDurationMS	389	ms
Averages	1	
AveragingMode	N.A.	
CoilElements	N.A. (synthetic)	
CombinedMetabolites	tNAA (NAA+NAAG), tCr (Cr+PCr), Cho (GPC+PCh), Ins (ml), Glx (Glu+Gln)	
CompressedSensingAccelerationFactor	N.A. (synthetic)	
DataProcessing	Model-synthesized signal (3D U-Net trained on real template subjects); no k-space reconstruction	
DeltaFrequencyPPM	0.0	ppm
Dimension	3D	
FIDPointsVectorSize	512	

FOV	220 x 220 x 130	mm
FlipAngle	45.0	°
MagneticFieldStrength	3.0	T
MatrixSize	44 x 44 x 25	
Measurements	1	
MetaboliteBasisSetLCModel	NAA, NAAG, Cr, PCr, GPC, PCh, ml, sl, Glu, Gln, Lac, GABA, GSH, Tau, Asp, Ala	
Note	SynthMRSI-Project is a synthetic dataset: real T1w anatomicals paired with model-synthesized MRSI metabolite signal (3D U-Net; see MRSIPrep manuscript, Validation Datasets). No real MRSI acquisition underlies these values -- numeric geometry/timing fields (TE, RepetitionTime, FOV, resolution, matrix size, field strength, etc.) record the acquisition protocol the synthetic signal generation was matched to; fields describing a specific real pulse sequence, reconstruction, or hardware are marked N.A. since none was used.	
Orientation	Transverse	
PhaseEncoding	N.A. (synthetic)	
PreparationScans	N.A. (synthetic)	
QualityMetrics	SNR, CRLB, FWHM (synthesized, not derived from spectral fitting)	
Quantification	N.A. (synthetic; not derived from spectral fitting)	
RFCoils	N.A. (synthetic)	
RemoveOversampling	N.A.	
RepetitionTime	0.457	s
ResolutionMM3	5 x 5 x 5.2	mm
RotationDeg	-0.79	°
SaturationBands	N.A. (synthetic)	
Scanner	N.A. (synthetic; no scanner)	
Sequence	N.A. (synthetic; not acquired with a real MRSI sequence)	
ShimMode	N.A. (synthetic)	
SlabThicknessMM	95	mm
Slabs	1	
SpectralBandwidthHz	1320	Hz
SpectralSuppr	N.A. (synthetic)	
TE	0.78	s
WaterReferenceFIDPoints	512	
WaterReferenceFlipAngle	45.0	°
WaterReferenceResolutionMM3	10.0 x 10.0 x 10.0	mm
WaterReferenceTE	0.72	s
WaterReferenceTR	0.46	s
WaterSuppr	N.A. (synthetic)	
WaterSupprBWHz	N.A. (synthetic)	Hz

PrepParams

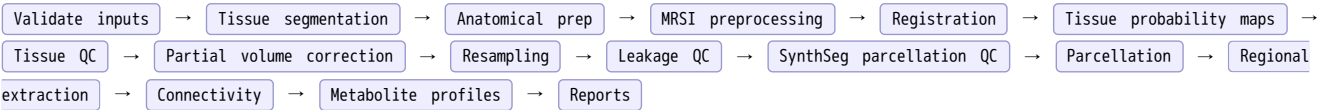
PrepParams

Processing parameters

Parameter	Value
Nucleus	1H
Tissue backend	synthseg-fast
Registration backend	ants
Registration T1w target	brain-csf

Normalization	simple
Output spaces	MNI152NLin2009cAsym
Parcellation mode	chimera
Atlas	chimera-LFMIHIFIS_scale3
SNR min	4.0
Linewidth max	0.1
CRLB max	20.0
Bi-harmonic spike filtering	True
Spike percentile	99.0
PVC disabled	False
T1 saturation correction	none
Connectivity	True
nproc x nthreads	1 x 12

Pipeline stages



Runtime

Runtime

Per-step duration

nproc: 1 | nthreads: 12

Excludes this report-generation step's own duration (not yet known while it's still running) and any time spent before this recording's pipeline started (e.g. queued behind other recordings under `--nproc`).

PROC computed this run · **REUSED** existing outputs reused (re-run with `--overwrite` to recompute) · **N/A** not requested by this configuration · **FAILED** raised an error

Step	Outcome	Duration	% of total
Tissue segmentation	PROC	56.6s	1.2%
Anatomical preparation	PROC	0.3s	0.0%
MRSI preprocessing	PROC	6.4s	0.1%
MRSI-to-T1w registration	PROC	1m 01.5s	1.3%
T1w-to-template registration	PROC	1m 43.2s	2.2%
Tissue probability maps in MRSI space	PROC	0.6s	0.0%
Partial volume correction	PROC	0.2s	0.0%
Resampling MRSI maps to T1w/MNI space	PROC	7.2s	0.2%
Signal leakage QC	PROC	0.0s	0.0%
SynthSeg parcellation and QC	REUSED	2.3s	0.0%
Parcellation	PROC	1h 13m 17.2s	94.6%
Regional metabolite extraction	PROC	3.3s	0.1%
Regional metabolic profiles and connectivity	PROC	8.8s	0.2%
T1 saturation correction --t1-correction none	N/A	-	-
Total (through report generation)		1h 17m 27.4s	100.0%

Outputs

Outputs

```
ses-01/
├── anat/
│   ├── synthseg/
│   │   ├── sub-01_ses-01_atlas-synthseg_desc-fastGMMW.tsv
│   │   ├── sub-01_ses-01_atlas-synthseg_desc-parcelqc.tsv
│   │   ├── sub-01_ses-01_space-mrsi_atlas-chimeraLFIHIFIFF_scale-scale3_desc-regional_metabolites.tsv
│   │   ├── sub-01_ses-01_space-mrsi_atlas-synthseg_desc-fastGMMW_dseg.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_atlas-synthseg_desc-fastGMMW_dseg.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_desc-brainCSF_T1w.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_desc-brainCSFmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_desc-synthsegBrain_mask.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_desc-synthsegBrain_T1w.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_desc-synthsegFastInput_T1w.nii.gz
│   │   └── sub-01_ses-01_space-T1w_desc-synthsegParcFast_dseg.nii.gz
│   ├── confounds/
│   │   ├── sub-01_ses-01_desc-leakageqc.tsv
│   │   ├── sub-01_ses-01_desc-mrsiqc.tsv
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_desc-fwhm_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_desc-snr_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-CrPCr_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-GluGln_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-GPCPCh_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-Ins_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-NAANAAG_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_desc-brain_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_label-CSF_probseg.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_label-GM_probseg.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_label-WM_probseg.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-qcmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-spikemask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-qcmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-spikemask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-qcmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-spikemask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-qcmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-spikemask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-qcmask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-spikemask_mask.nii.gz
│   │   ├── sub-01_ses-01_space-orig_desc-fwhm_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_desc-snr_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_met-CrPCr_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_met-GluGln_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_met-GPCPCh_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_met-Ins_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-orig_met-NAANAAG_desc-crlb_mrsi.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_label-CSF_probseg.nii.gz
│   │   ├── sub-01_ses-01_space-T1w_label-GM_probseg.nii.gz
│   │   └── sub-01_ses-01_space-T1w_label-WM_probseg.nii.gz
│   ├── connectivity/
│   │   ├── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scale3_npert-50_filt-biharmonic_pvcrr_GM_desc-connectivity_mrsi.npz
│   │   ├── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scale3_npert-50_filt-biharmonic_pvcrr_GM_desc-connectivity_mrsi.tsv
│   │   ├── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scale3_npert-50_filt-biharmonic_pvcrr_GM_desc-edges_mrsi.tsv
│   │   ├── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scale3_npert-50_filt-biharmonic_pvcrr_GM_desc-metabolicprofiles_mrsi.npz
│   │   └── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scale3_npert-50_filt-biharmonic_pvcrr_GM_desc-nodes_mrsi.tsv
│   ├── logs/
│   │   └── sub-01_ses-01_desc-mrsiprep_log.txt
│   ├── mrsi/
│   │   ├── mni/
│   │   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-CrPCr_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-GluGln_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-GPCPCh_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-Ins_desc-signal_mrsi.nii.gz
│   │   │   └── sub-01_ses-01_space-MNI152Nlin2009cAsym_res-5_met-NAANAAG_desc-signal_mrsi.nii.gz
│   │   ├── orig/
│   │   │   ├── sub-01_ses-01_space-mrsi_desc-reference_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signalspikefilt_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signalspikefilt_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signalspikefilt_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signal_mrsi.nii.gz
│   │   │   ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signalspikefilt_mrsi.nii.gz
│   │   │   └── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signal_mrsi.nii.gz
```

```
|
|   └── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signalspikefilt_mrsi.nii.gz
|   └── orig-pvc/
|       ├── sub-01_ses-01_space-mrsi_met-CrPCr_desc-signalpvc_mrsi.nii.gz
|       ├── sub-01_ses-01_space-mrsi_met-GluGln_desc-signalpvc_mrsi.nii.gz
|       ├── sub-01_ses-01_space-mrsi_met-GPCPCh_desc-signalpvc_mrsi.nii.gz
|       ├── sub-01_ses-01_space-mrsi_met-Ins_desc-signalpvc_mrsi.nii.gz
|       └── sub-01_ses-01_space-mrsi_met-NAANAAG_desc-signalpvc_mrsi.nii.gz
|   └── parcel/
|       └── sub-01_ses-01_atlas-chimeraLFIHIFIFF_scalescale3_desc-GMmetprofiles_mrsi.npz
└── transforms/
    ├── anat/
    │   ├── sub-01_ses-01_desc-t1w_to_mni.affine.mat
    │   ├── sub-01_ses-01_desc-t1w_to_mni.affine_inv.mat
    │   ├── sub-01_ses-01_desc-t1w_to_mni.syn.nii.gz
    │   └── sub-01_ses-01_desc-t1w_to_mni.syn_inv.nii.gz
    └── mrsi/
        ├── sub-01_ses-01_desc-mrsi_to_t1w.affine.mat
        ├── sub-01_ses-01_desc-mrsi_to_t1w.affine_inv.mat
        ├── sub-01_ses-01_desc-mrsi_to_t1w.syn.nii.gz
        └── sub-01_ses-01_desc-mrsi_to_t1w.syn_inv.nii.gz
```

Citations

MRSIPrep: see CITATION.cff in the [MRSIPrep repository](#) for how to cite this software.

MRSI acquisition reporting follows the [MRSinMRS](#) minimum reporting standard (Lin et al. 2021).